

Blockchain y sus aplicaciones

DeFi (Finanzas Descentralizadas)

Gastón Escobar - 2026



Universidad
de la Ciudad
de Buenos Aires

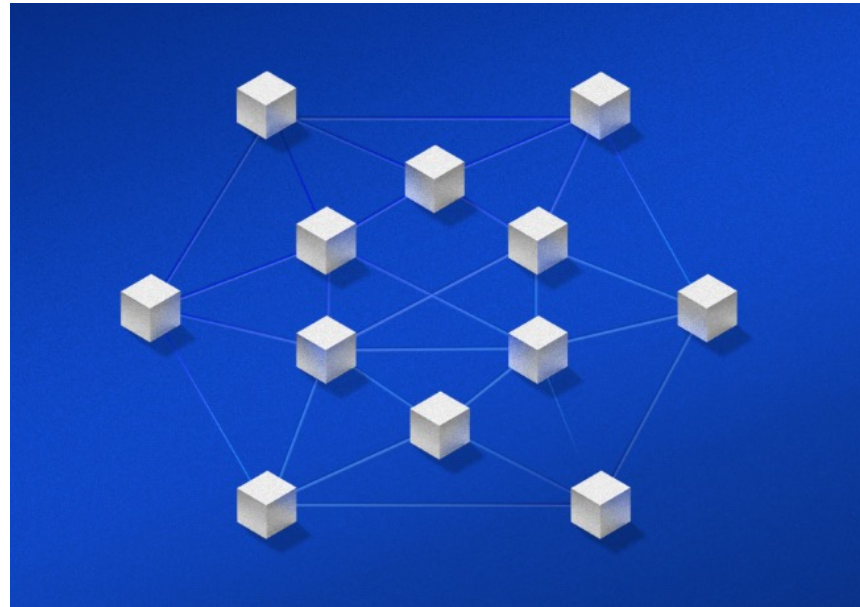
DEFI



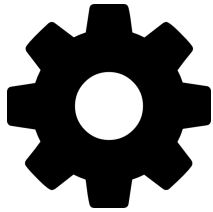
stón Escobar

Definición de Blockchain

“Es una base de datos PÚBLICA que registra transacciones digitales INMUTABLES de forma DESCENTRALIZADA y sin necesidad de un TERCERO DE CONFIANZA, que representa una verdad ÚNICA, GLOBAL y CONSENSUADA”



Tecnología Blockchain



TECNOLOGÍA

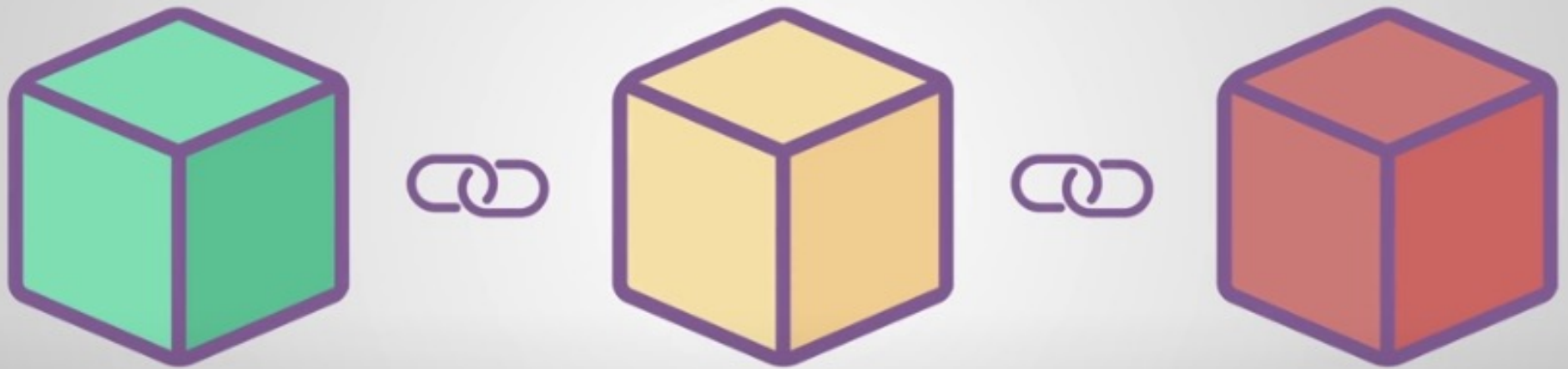


BASE DE DATOS

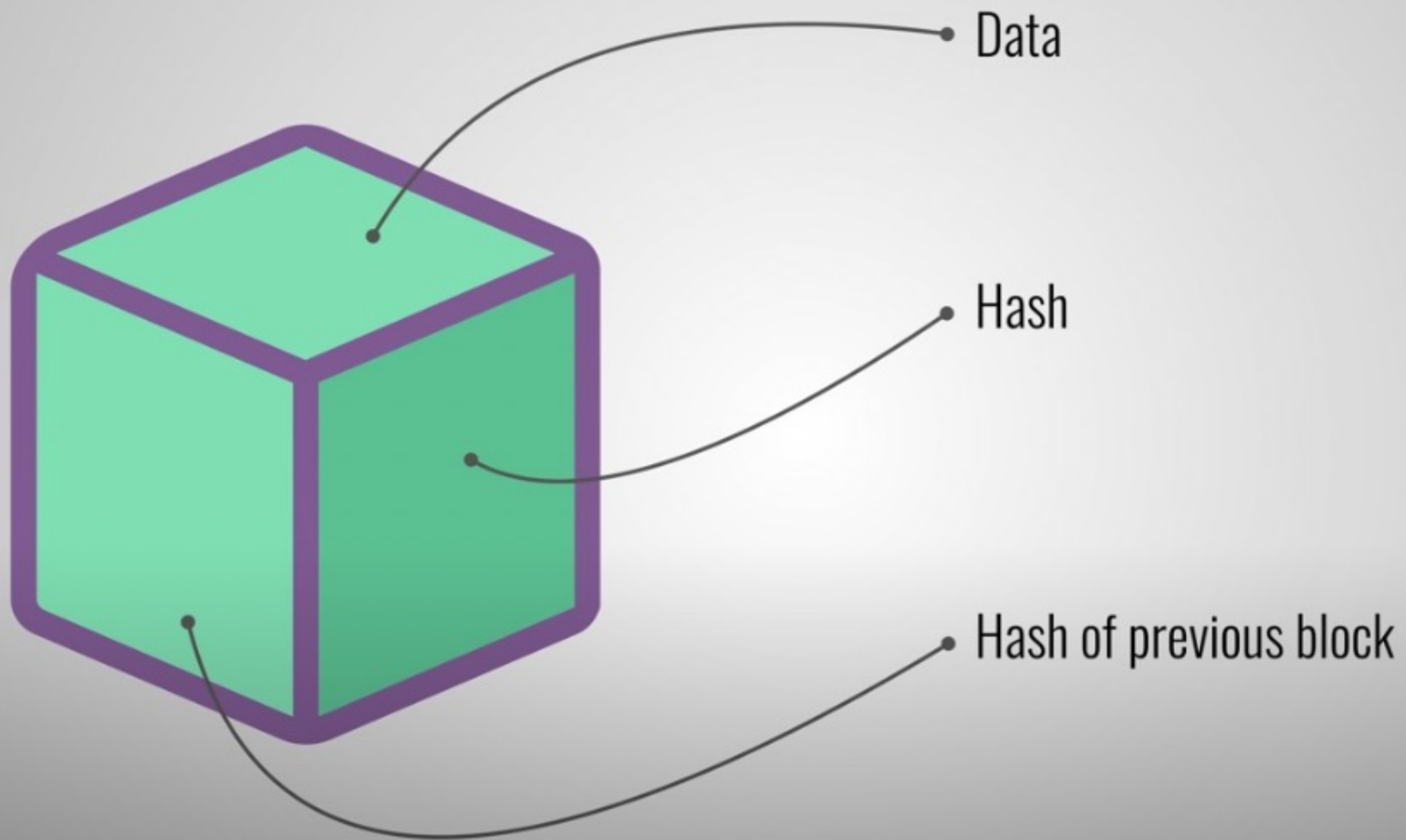


RED

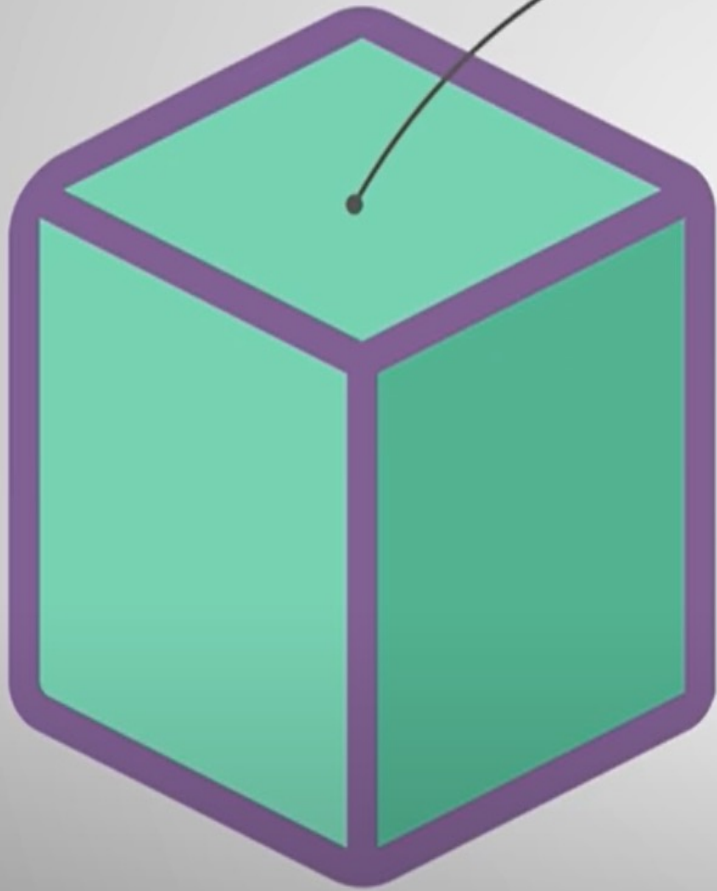
Blockchain: ¿cómo funciona la cadena de bloques?



Distributed ledger



Data



From:



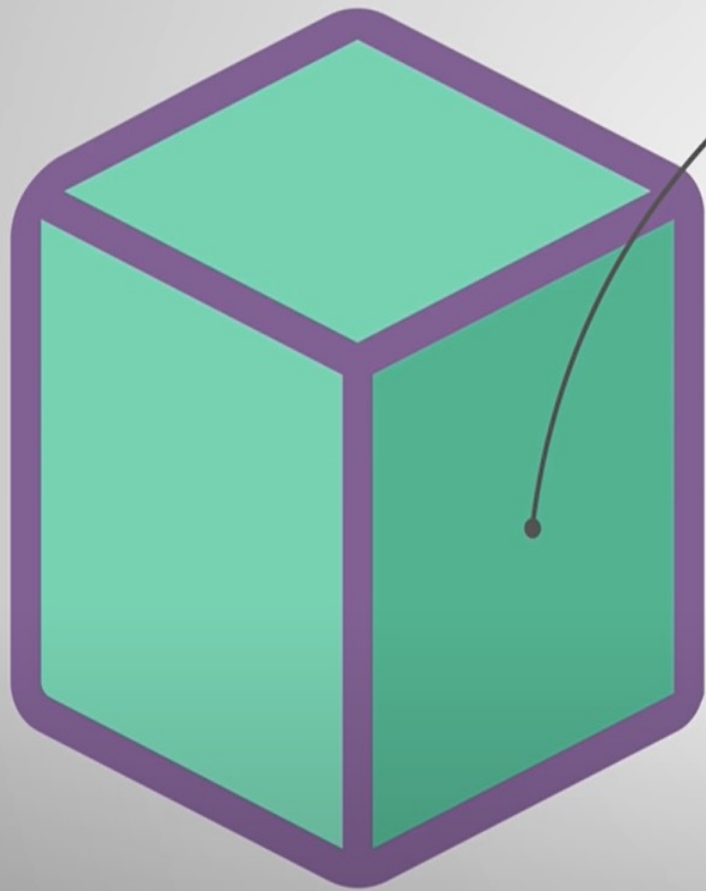
To:



Amount:



Bitcoin block example

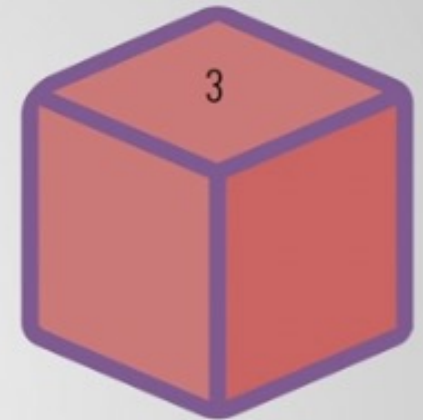
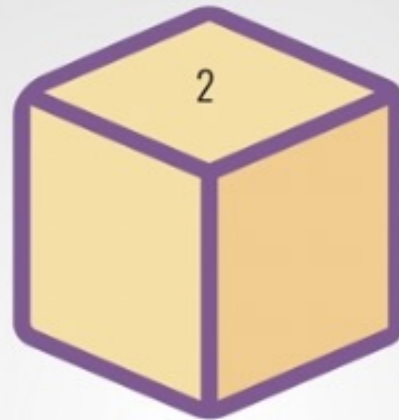
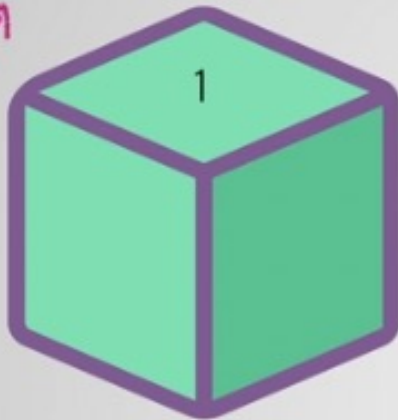


Hash

e2c521bc53bb5db4fc0aa497da2ba5d4c8444db3



Genesis block



Hash:

1Z8F

Hash:

6BQ1

Hash:

3H4Q

Previous hash:

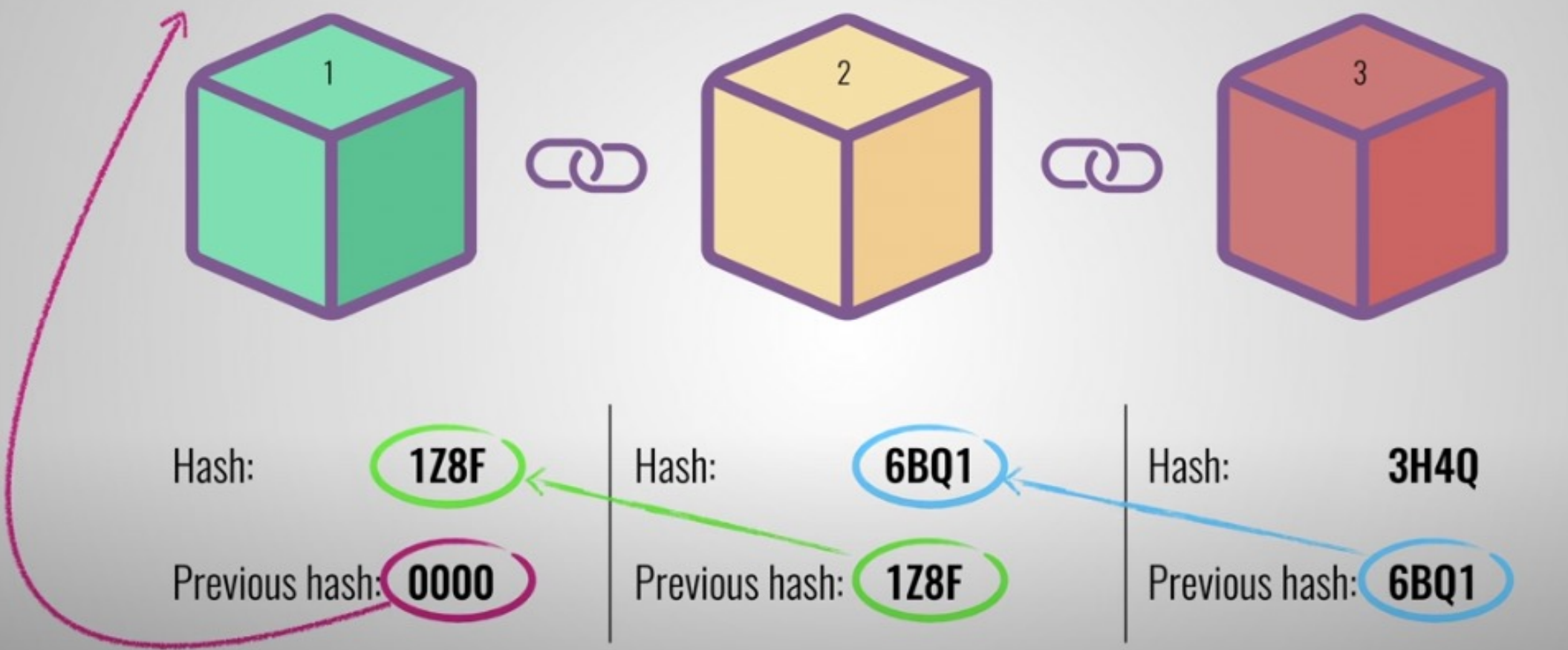
0000

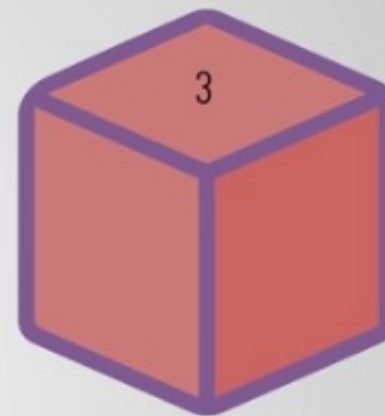
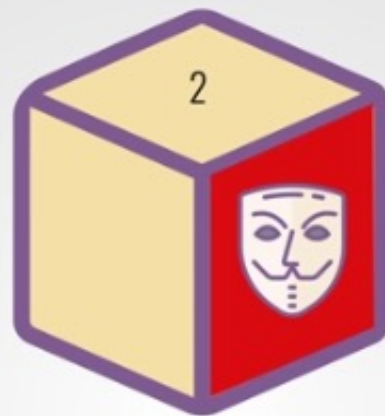
Previous hash:

1Z8F

Previous hash:

6BQ1





Hash: **1Z8F**

Previous hash: **0000**

Hash: ~~6BQ1~~ **H62Y**

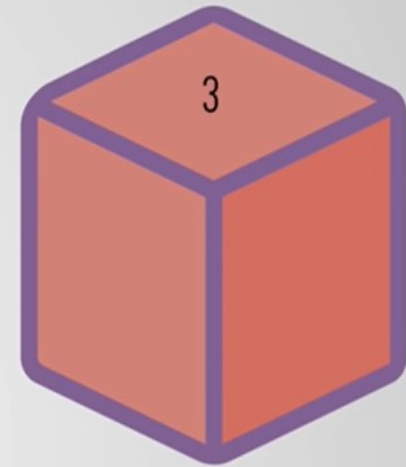
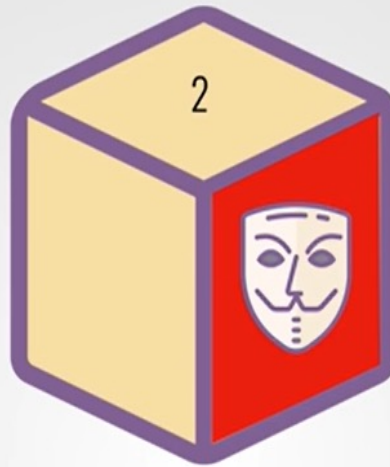
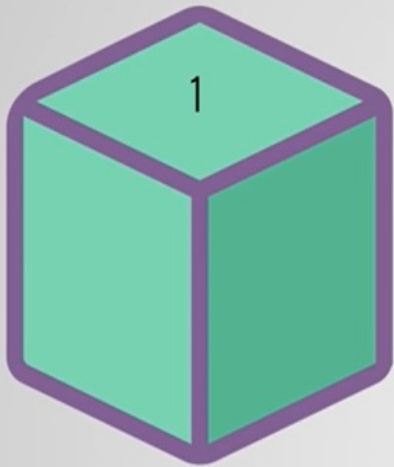
Previous hash: **1Z8F**

Hash: **3H4Q**

Previous hash: **6BQ1**

Uh thats
not right??

Proof of Work



10 minutes



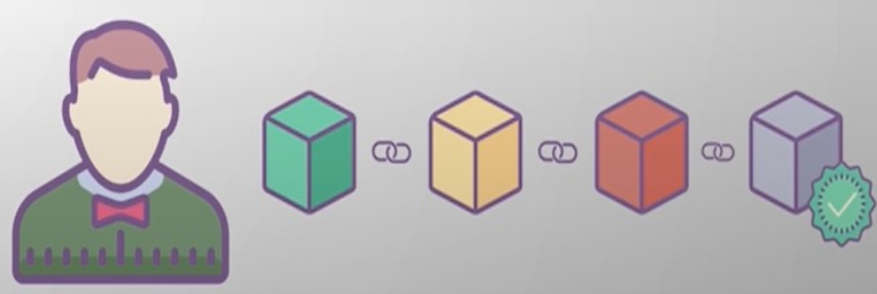
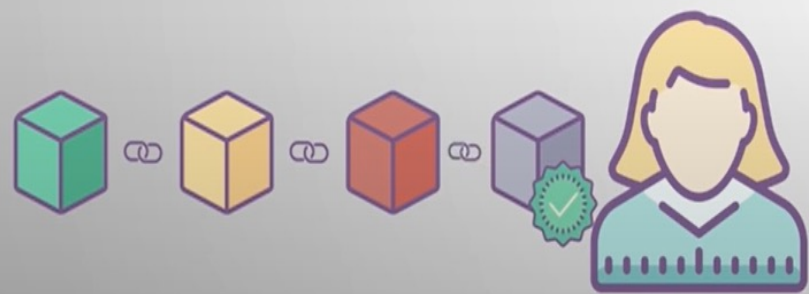
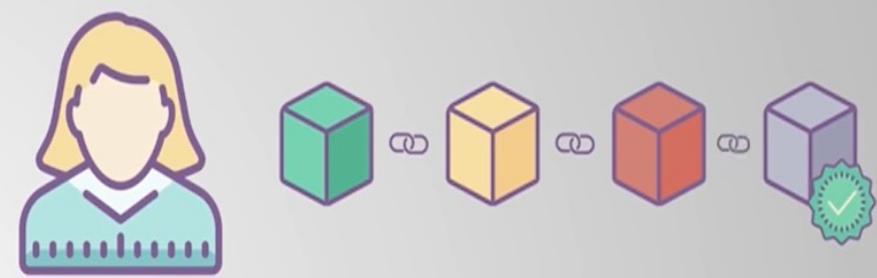
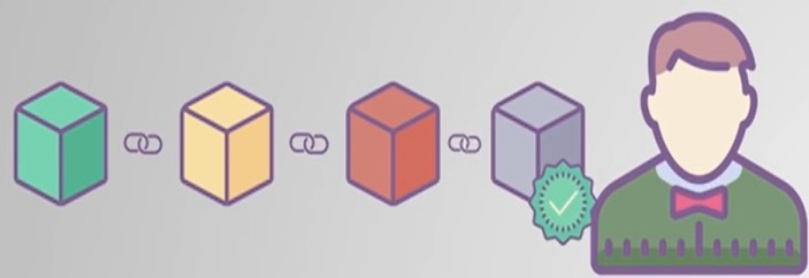
10 minutes

Distribuida



P2P Network

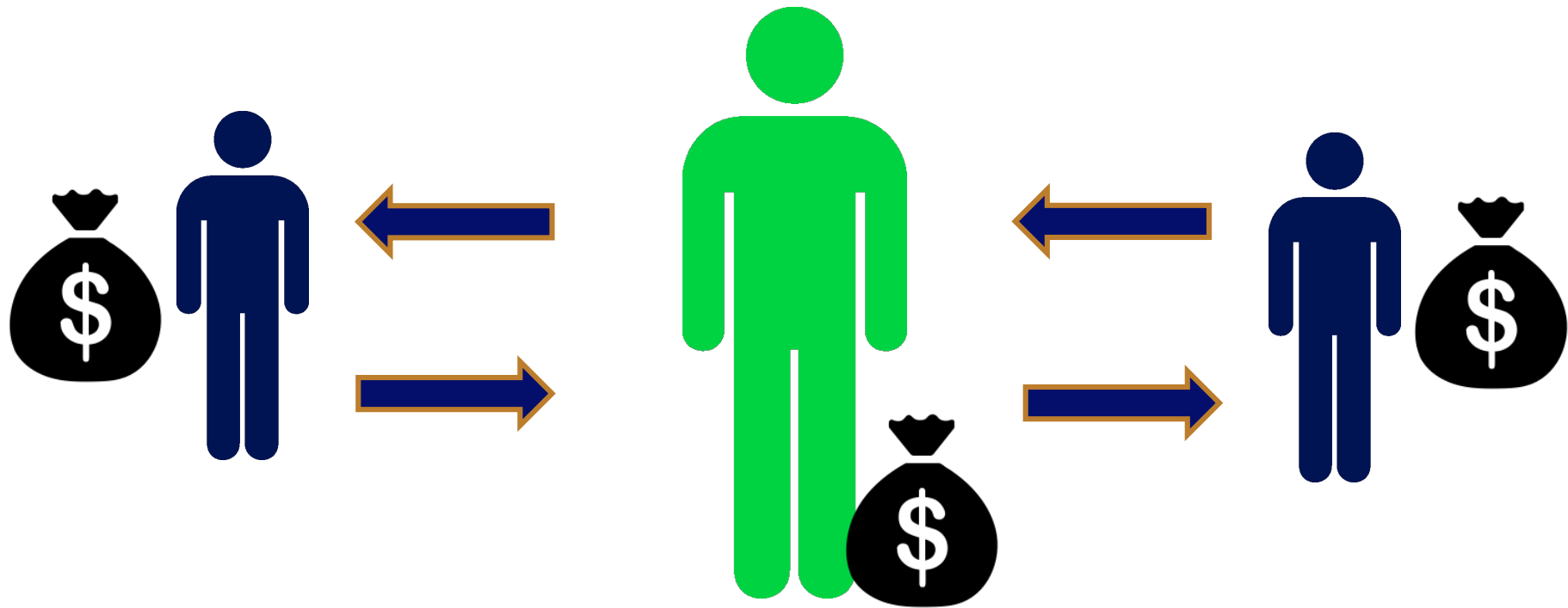




CARACTERÍSTICAS

- 1.Descentralización**
- 2.Inmutabilidad**
- 3.Transparencia**
- 4.Seguridad Criptográfica**
- 5.Consenso**
- 6.Anonimato o Pseudonimato**
- 7.Escalabilidad (en desarrollo)**

PROTOCOS DE CONSENSO



CONFIANZA
CENTRALIZADA

PROTOSCOLOS DE CONCENSO



**CONFIANZA
DESCENTRALIZADA**

PROTOCOS DE CONSENSO

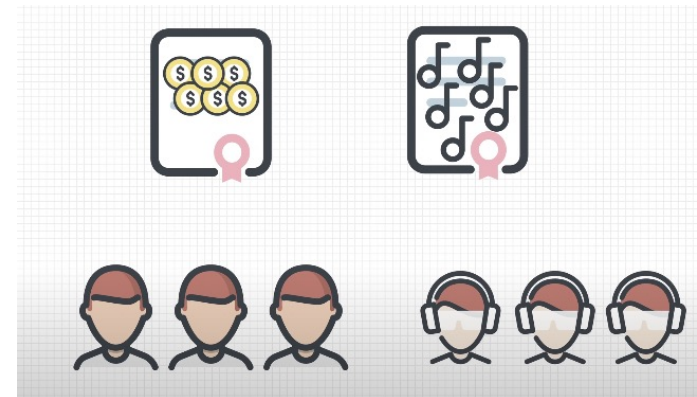
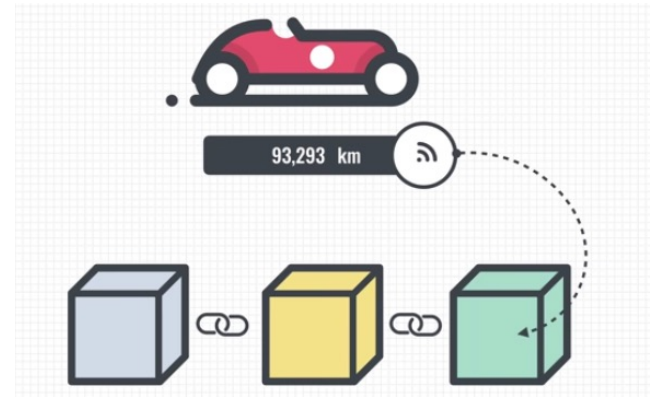


PROTOSCOLOS DE CONCENSO



Aplicaciones

- Cryptocurrencies
- Kilometraje de los autos
- Escribanía
- Voto Digital
- Industria alimenticia
- Trackear paquetes
- Historial medico digital
- Personal ID
- Colectar royalties a artistas



¿Qué es DeFi?

“Servicios Financieros sin Bancos”

¿Una Fintech
es DeFi??

DINERO

CARACTERÍSTICAS

- ESCASO
- DURABLE
- TRANSPORTABLE
- DIVISIBLE
- RECONOCIBLE
- FUNGIBLE



FUNCIONES

1. RESERVA DE VALOR
2. MEDIO DE CAMBIO
3. UNIDAD DE CUENTA

Comparación

	 GOLD	 BITCOIN	 FIAT CURRENCY	
 DURABLE	+	+	—	While all are physically durable, fiat currency over history has not maintained purchasing power durability
 DIVISIBLE	—	+	+	Physical gold is only divisible to small pieces; bitcoin is divisible to eight decimals
 FUNGIBLE	+	+	—	Gold and bitcoin are fungible, but fiat currency is not fungible with other fiat (US Dollar is not fungible with Canadian dollar)
 PORTABLE	—	+	+	Gold has a high value to weight ratio, but compared to the others is still heavy and cumbersome to transport
 VERIFIABLE	—	+	—	Both gold and fiat currency have been counterfeited; gold can be verified but only through cumbersome assay
 SCARCE	+	+	—	Gold is scarce, bitcoin is scarce and finite; the only constraint on fiat currency is willingness of government or central bank
 TRACK RECORD	+	—	—	Gold has the longest track record as money and maintaining purchasing power; bitcoin's history is the shortest; fiat currency has a poor track record ²

EVOLUTION OF MONEY



BARTER



GOLD



**METALL
COINS**



**PAPER
MONEY**



**PLASTIC
CARDS**



**ELECTRONIC
MONEY**



**CRYPTO
CURRENCY**

Principales Blockchain y criptomonedas

BLOCKCHAIN



REDES TRANSACCIONALES (LAYER 1)

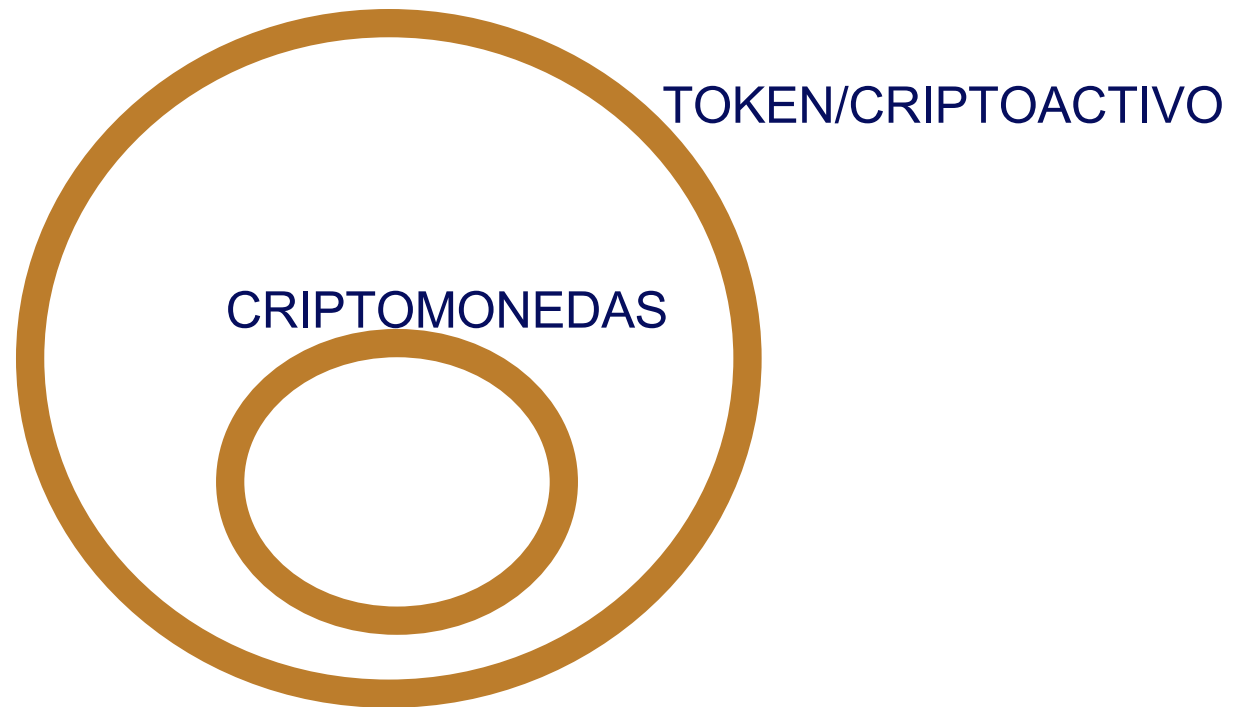


CRIPTOMONEDA (Dinero) EN CADA RED:
BTC, ETH, ADA, SOL, BNB

Criptomonedas y tokens

TOKEN (CRIPTOACTIVO)= ACTIVO DIGITAL DENTRO DE LA BLOCKCHAIN

CRIPATOMONEDA = TOKEN NATIVO DE CADA BLOCKCHAIN



Smart Contracts



BITCOIN 



ETHEREUM 

AI + Blockchain: La Convergencia

- Dos tecnologías que se potencian mutuamente
- AI necesita datos confiables → Blockchain provee datos inmutables y verificables
- Blockchain necesita automatización inteligente → AI optimiza contratos y protocolos
- Juntas habilitan: agentes autónomos, mercados predictivos, gobernanza inteligente

AI Aplicada a DeFi: Casos de Uso

- **AI Agents para trading automatizado (ej: Autonolas, AI16z)**
- **Detección de fraude y anomalías en tiempo real con ML**
- **Scoring crediticio descentralizado sin datos personales**
- **Optimización automática de yield farming con modelos predictivos**
- **Auditoría de smart contracts con LLMs (ej: Slither + GPT)**
- **Oráculos inteligentes que validan datos off-chain con AI**

Muchas gracias